

Shihabul Ehsan

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EDUCATION

B.Sc. in Electrical and Electronic Engineering

Jan 2020 – Sep 2025

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

CGPA: 3.25 / 4.00

Thesis: Design and Development of a Low-cost Dual-core Flight Controller for Quadcopter and Testing its Performance under Partial Propeller Disturbance Condition.

Achievements: Helped more than 100 students learn the basics of hardware and robotics by re-establishing the Robotics Club of KUET.

RESEARCH INTERESTS

Embedded Systems, Control Systems, Sensor Fusion, Robotics, UAV, ROS 2, Machine Learning, Deep Learning

PCB DESIGN EXPERIENCE

Remote Freelance PCB Layout Engineer

Dec 2025 – Present

Balboa Inspection & Maintenance, LLC · Contract (hourly)

Altium Designer

- Laid out a **1 Gbps 7-port Ethernet switch** with 5 integrated PHYs and 2 external PHYs over RGMII, using length-matched high-speed routing; manufactured and deployed in Balboa's Copperhead inspection robot.
- Laid out a **3-board Altium multi-board system** stacked inside a cylindrical enclosure, with dense component placement:
 - 300 V DC to 24 V DC isolated converter**
 - Power converter:** 24 V to 5 V, then 3.3, 2.5, 1.8, 1.2 and 1.1 V rails
 - Optical-to-Ethernet converter:** 2 optical + 2 copper ports, Ethernet switch IC and 2 PHYs
- Worked from client-supplied schematics in Altium Designer.

PROJECTS

Vector Drive V1: 3-Board BLDC Motor Drive System ([images](#))

- Designed the schematic and PCB layout of a 3-board system that drives two BLDC motors as the wheels of a high-speed ground robot.
- Rendered with PCB Studio, my own Blender extension ([below](#)).

PCB Studio: Blender Extension for Realistic PCB Renders ([demo](#))

- Built a Blender extension in Python that turns Altium PCB exports (OBJ + MTL) into realistic product renders and turntable animations, for engineers with no Blender experience.
- One-click scene setup with PCB material presets, studio lighting, camera presets and Cycles/EEVEE rendering; released v2.3.2, tested on Blender 4.5 and 5.2 LTS ([GitHub](#)).

Low-Power Cattle Health Monitor (client project) ([images](#))

- Designed the schematic and PCB in KiCad for an ESP32-C6 device that tracks cattle body orientation (accelerometer) and body temperature; firmware by another developer.
- Low-power design: 450 μ A average current (calculated), about 6 months on one charge of a 1S 2200 mAh Li-ion battery; manufactured by PCBWay.

ESP32 Mini Drone with PCB Frame ([images](#))

- Designed the schematic and PCB in KiCad; the PCB doubles as the airframe, with the arms and motor mounts built into the board. Inspired by Espressif's ESP-Drone.
- Chose a BMI088 IMU over the MPU6050 after comparing both in real-world tests; manufactured by JLCPCB in framed and frameless versions, and it flies.
- An earlier EasyEDA prototype with an MPU6050, milled in the KUET fabrication lab ([link](#)), also flew ([video](#)).

Smartphone-Controlled Mini Plane: ESP8266 Flight Controller PCBs ([flight demo](#))

- Designed two ESP8266 flight controller PCBs in KiCad:
 - No-IMU version** ([images](#)): two unidirectional DC motor drivers, flown over Wi-Fi from a phone; the self-stable airframe glides without stabilization.
 - BMI088 IMU version** ([images](#)): four motor outputs; PCB sponsored by JLCPCB.
- Implemented a PID roll stabilizer, first tested with an MPU6050.

Advanced Line Follower Robot ([link](#))

- Built more than 10 line follower robots.
- Participated in 8 national line follower robot competitions ([link](#)).
- Still working on bringing this product to market for kids ([link](#)).

Pipeline Corrosion Monitoring Robot ([link](#))

- Designed and developed the circuit and code of the vehicle.

Humanoid Eye Toy that can Interact with Kids ([link](#))

- This is my third-year project.

Flight Controller for Quadcopter from Scratch ([link](#))

- This is my thesis project.

Radio Controlled Glider Plane ([link](#))

- Implemented my quadcopter flight controller on this foam plane.

Development of an Extended Kalman Filter for Low-Cost Quadcopter

- Developed the extended Kalman filter in Python; not yet implemented on the quadcopter.
- Also developed an unscented Kalman filter for 6-DOF and 9-DOF IMUs.

SKILLS

Programming Languages: C, C++, Python, MATLAB

PCB Design: Altium Designer, KiCad **ADVANCED** · EasyEDA **FAMILIAR**

3D Design: Fusion 360

Microcontrollers: Arduino, ESP32, STM32, Raspberry Pi

ML/DL Frameworks: TensorFlow, PyTorch, OpenCV

Robotics: ROS 2

LANGUAGES

English: Fluent

Bangla: Fluent

REFERENCES

Dr. Md. Shahjahan

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